Mahmoud Alyosify

LLM & Generative AI Engineer | Fine-Tuning, RAG & Inference Optimisation
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PROFESSIONAL SUMMARY

LLM & Generative AI Engineer specialising in **RAG, agentic LLM workflows, fine-tuning and inference-time optimisation** of open-weight language models. Currently an **AI Engineer Intern at Injaz Tech**, building an agentic RAG assistant with **LangChain, LangGraph, FAISS and FastAPI**. Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**; master's thesis (**MinimalLM**) is a training-free decoding method that makes **Mistral, Llama-3.1 and Qwen2.5** generate **47.6% fewer tokens** with no answer cut off. Hands-on with **LoRA/QLoRA fine-tuning at 159K-record scale**, self-hosted inference, LLM evaluation (BERTScore, ROUGE, LLM-as-Judge) and AWS GPU deployment. Focused on making language-model systems **cheaper, faster and measurably reliable** in production.

EDUCATION

MSc in Artificial Intelligence — *Sep 2025 – Nov 2026*
Queen's University, Kingston, Ontario, Canada — Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- **Master's Thesis:** *MinimalLM: Verbosity-Aware Decoding for Token-Efficient Language Model Inference* — Supervisor: Dr. Rahatara Ferdousi.
- Coursework: **Large Language Models, Generative AI, Deep Learning, Cloud Computing, End-to-End ML Pipelines**.

B.Sc. Computer & Information Science (Bioinformatics) — *Sep 2019 – Jul 2023*
Assiut University, Egypt — GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

TECHNICAL SKILLS

LLM & Generative AI	Large Language Models, Instruction Fine-Tuning, LoRA, QLoRA, PEFT, RAG, Agentic RAG (Routing, Relevance Grading, Query Rewriting, Self-Checks), Hybrid Search (BM25 + Dense, RRF), Cross-Encoder Re-Ranking, Custom Decoding Strategies, LogitsProcessors, Sampling & Entropy-Based Gating, Prompt Engineering, Structured Outputs, Tool / Function Calling, Multi-Agent Systems
Frameworks & Serving	PyTorch, Hugging Face Transformers, PEFT, Datasets, Accelerate, LangChain, LangGraph, Ollama, Pydantic, FAISS, ChromaDB, ONNX, FastAPI, Streamlit, Self-Hosted Inference, Docker, NVIDIA A40 GPU, AWS
Models Worked With	Mistral-7B-Instruct, Llama-3.1-8B-Instruct, Meta-Llama-3-8B, Qwen2.5-3B-Instruct
LLM Evaluation	MT-Bench, AlpacaEval, Arena-Hard-Auto, BERTScore, ROUGE, LLM-as-Judge (Judge Validation), RAG Groundedness & Retrieval Metrics (Top-1, MRR), CI Evaluation Gates, Non-Inferiority Testing, Ablation Studies, Deterministic & Reproducible Benchmarking
Efficiency & Systems	Token-Cost Reduction, Per-Token & Per-Answer Latency Measurement, GPU-Time Accounting, KV-Cache-Free Methods, Quantised Fine-Tuning (4-bit)
Programming	Python, SQL, C++, C#, JavaScript, Bash, Git/GitHub, Linux

PROFESSIONAL EXPERIENCE

AI Engineer Intern — *Oct 2026 – Present*
Injaz Tech — Injaz Group (formerly Fingerprint Consultancy) — Maadi, Cairo, Egypt

- Built an **agentic RAG** assistant (**LangChain, LangGraph**, FAISS, cross-encoder re-ranking) served via **FastAPI** with **Docker**, **136 pytest** tests and a CI evaluation gate.
- Raised top-1 retrieval accuracy from **75% to 94%**; a redesigned relevance grader lifted recall from **17% to 88%** and cut false refusals from **5/20 to 1/20**.

Part-Time Instructor — Machine Learning & AI for Business — *Jul 2025 – Present*
National Telecommunication Institute (NTI) — Egypt

- Teach **Machine Learning** (90-hour bootcamps) and **AI for Business** modules in NTI's Summer Internships Program, on separate engagements.

AI in Cybersecurity Intern — *Jun 2025 – Jul 2025*
e& Egypt — via NTI (On-the-Job Training) — Cairo, Egypt

- Applied AI-driven tooling across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) *Feb 2024 – Mar 2025*
National Company for Roads Building & Development Cairo, Egypt

- Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles**; co-developed an internal data-management application in **C#, .NET, SQL Server**.

Online Instructor — AI, Algorithms & Data Science *2020 – Present*
Udemy & YouTube Remote

- Built Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

SELECTED LLM RESEARCH & ENGINEERING PROJECTS

MinimalLM: Verbosity-Aware Decoding for Token-Efficient Language Model Inference *2026*
MSc Thesis, Queen's University · Supervisor: Dr. Rahatara Ferdousi

- Fewer generated tokens are a real saving only when the answer is still complete. **MinimalLM gives both, without training.** A guardrail system prompt, an entropy-gated stop-token **LogitsProcessor**, a completeness gate and a final check make unchanged **Mistral-7B, Llama-3.1-8B and Qwen2.5-3B** generate **47.6% fewer tokens** and answer **19–68% faster** on 330 questions (MT-Bench, AlpacaEval, Arena-Hard-Auto), with **no answer cut off**.
- LLM-judge score **8.04/10** against 8.42 for the unmodified models (cutting the originals to the same length: 6.46); **45.6%** fewer tokens in six other languages. The stop-token boost alone cuts off **59.3%** of its answers, which is why the instruction and the completeness gate are both needed. Cost: shorter answers more often leave out part of a broad question.

Horus-OSINT — Domain-Adapted LLM Assistant on AWS *2026*
Meta-Llama-3-8B · Instruction Fine-Tuning · AWS · Python

- Fine-tuned **Meta-Llama-3-8B** on **159,826** curated GTD/GDELT records, building the full pipeline from dataset construction and instruction formatting through training to evaluation.
- Deployed the adapted model on **AWS** as an interactive assistant that answers natural-language questions over global event and threat-intelligence data.

HORUS Sentinel — Multi-Agent LLM Platform over a Knowledge Graph *2026*
Multi-Agent Systems · Fine-Tuned LLM · Knowledge Graphs · Self-Hosted Inference

- Architected a swarm of specialised agents that continuously collect open-source data and correlate every finding into a **living knowledge graph** used as the retrieval substrate for generation.
- A **self-hosted, fine-tuned language model** reasons over that graph to produce **prioritised, evidence-backed reports**, grounding every claim in retrieved source material.

Automated Multimodal Agent — PDF to Narrated Presentation *2026*
Generative AI · Agentic Orchestration · Multimodal Generation

- Built an autonomous think–plan–act workflow that converts PDFs into fully narrated PowerPoint decks by chaining document understanding, structured slide generation and audio synthesis.

Additional Relevant Work

- **Human vs. Agentic Pull Requests** — reproducible GitHub-mining study quantifying review effort and failure modes of LLM-generated versus human-authored code.
- **SimCLR-Vision-SSL** — contrastive representation learning with **ONNX/FAISS** embedding retrieval — transferable to RAG vector-search pipelines.

CERTIFICATIONS & AWARDS

Generative AI & Prompt Engineering — IBM / Coursera

Machine Learning Specialization — DeepLearning.AI

Digilians Presidential Scholarship — Queen's University

1st Place — Smart Cities Hackathon 2022

AWS Academy — ML for NLP, ML Foundations, Data Engineering

HCIP – AI (Huawei)

Semi-Finalist — Microsoft Imagine Cup 2023

Ideal Student Award — Assiut University, 2022–2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency.